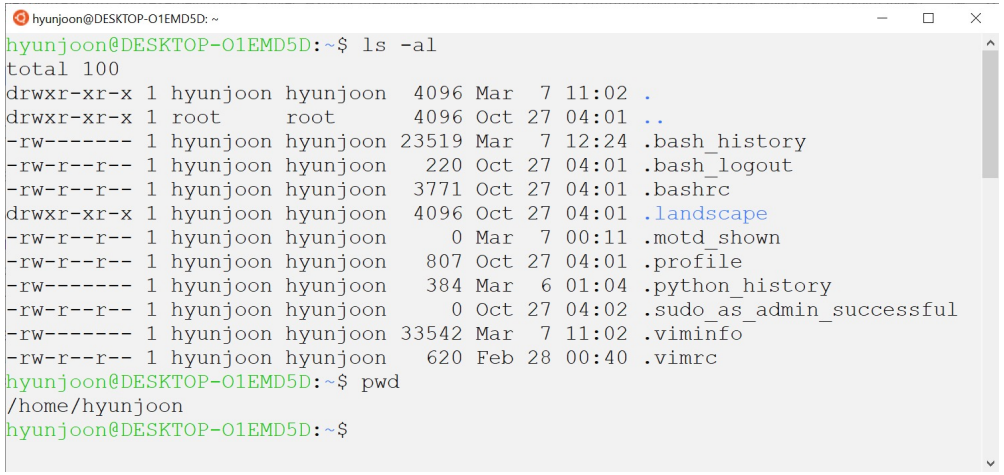
Practice 1. Setup

[CSE2010] Data Structures

Department of Data Science

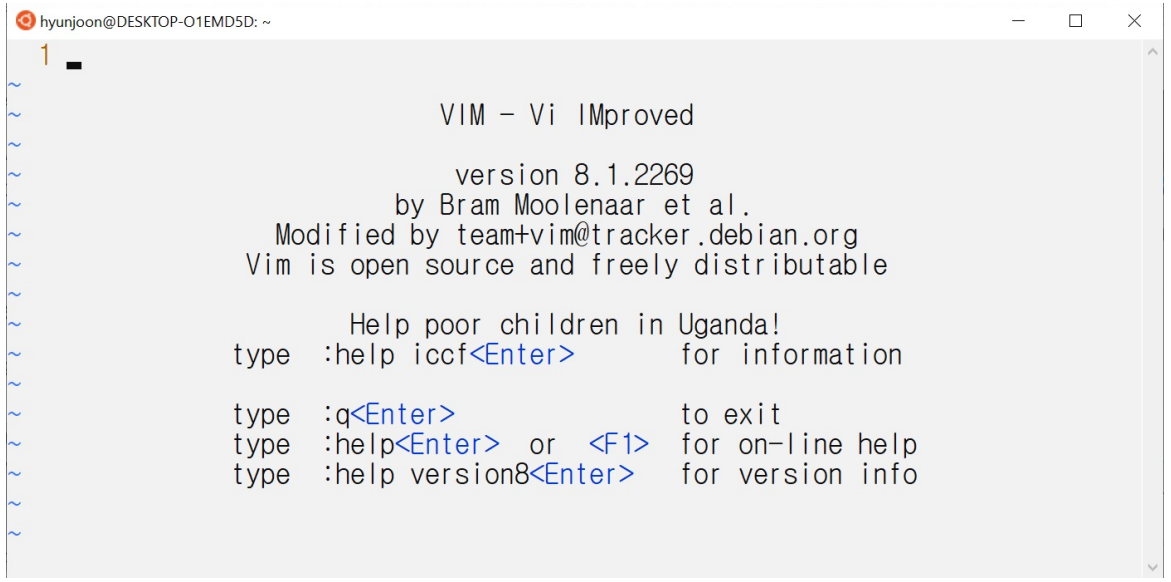
Linux

- Family of open-source Unix operating systems, based on Linux Kernel
  - Ubuntu, Fedora, Debian, CentOS, etc.
- In Linux, you interact with OS via a shell .
  - Shell: an interface that enables you to access the OS services
- Command-line interface (CLI) is mainly used.



Setup for Practices and Assignments

- Applications
  - MAC OS or Linux: Terminal
  - Windows 10: Ubuntu in Windows Subsystem for Linux (WSL)
- Editors
  - Vim
- Languages
  - C/C++
  - Python

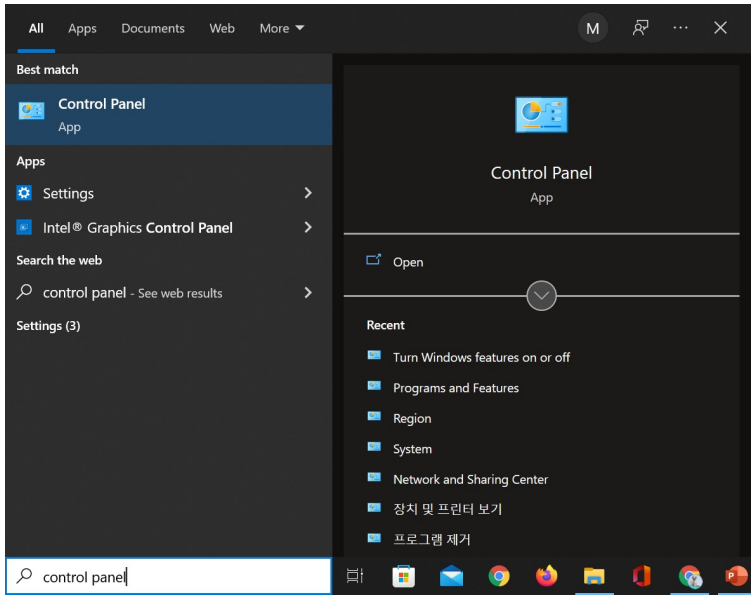


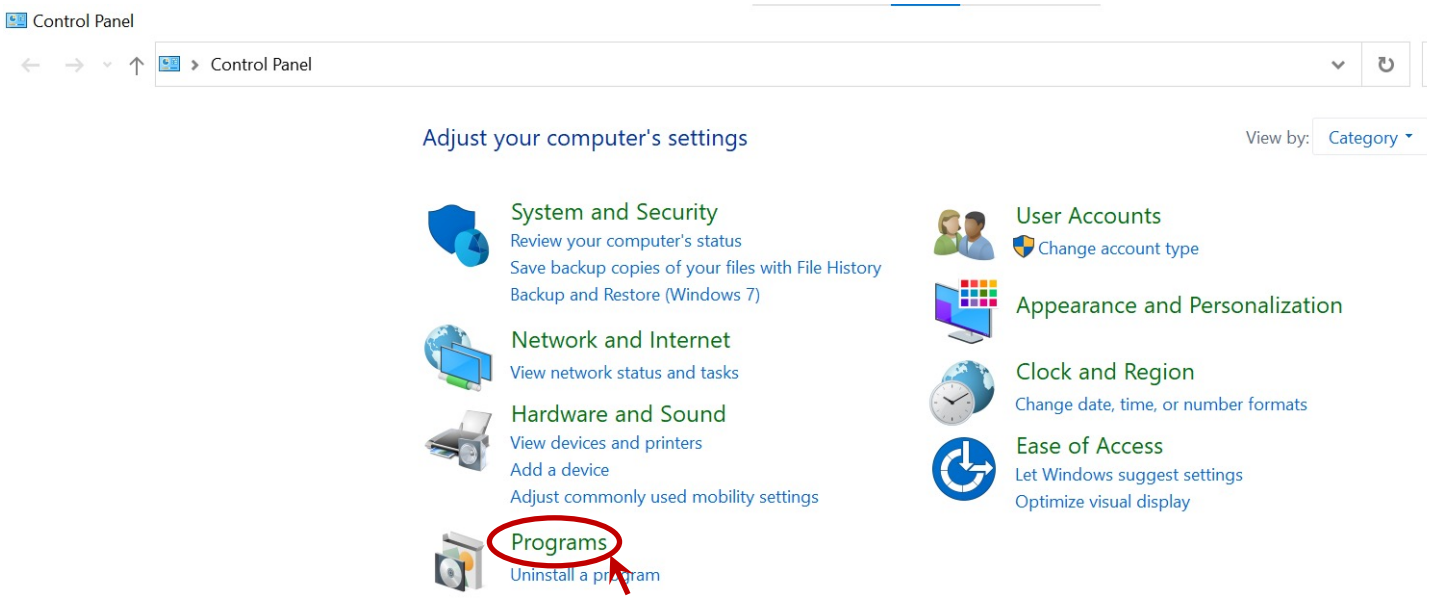
Setup for Windows 10 Users

1. Download and install Ubuntu from Microsoft Store.
   - https://m.blog.naver.com/elepartsblog/222205998011
2. Run Ubuntu in the Start menu.

Installing Ubuntu in Windows 10

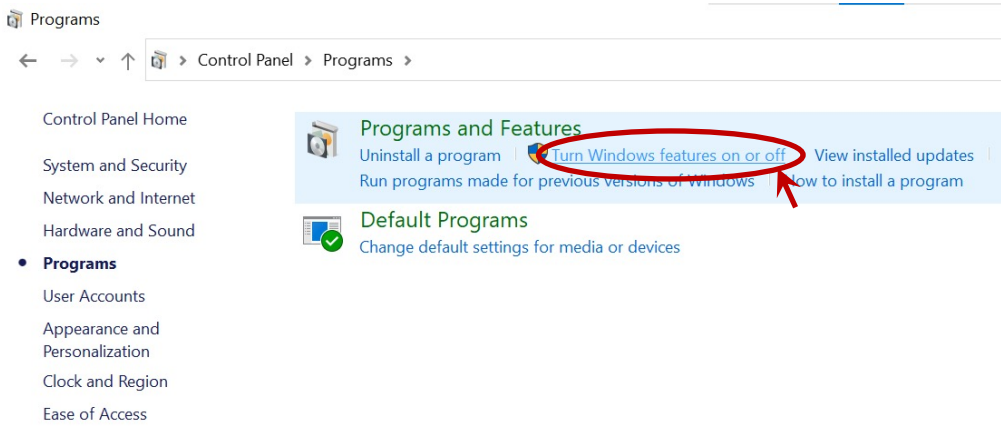
- Start menu → Control Panel → Programs

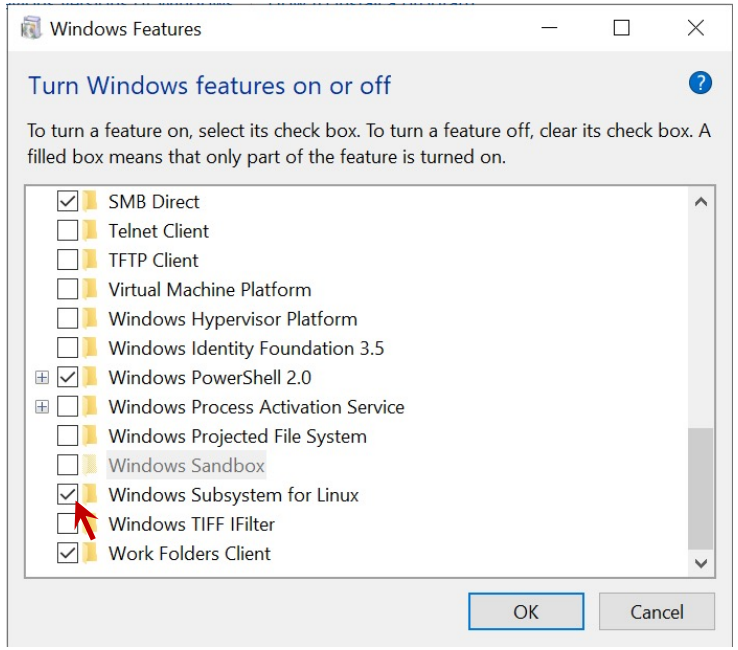




Installing Ubuntu in Windows 10

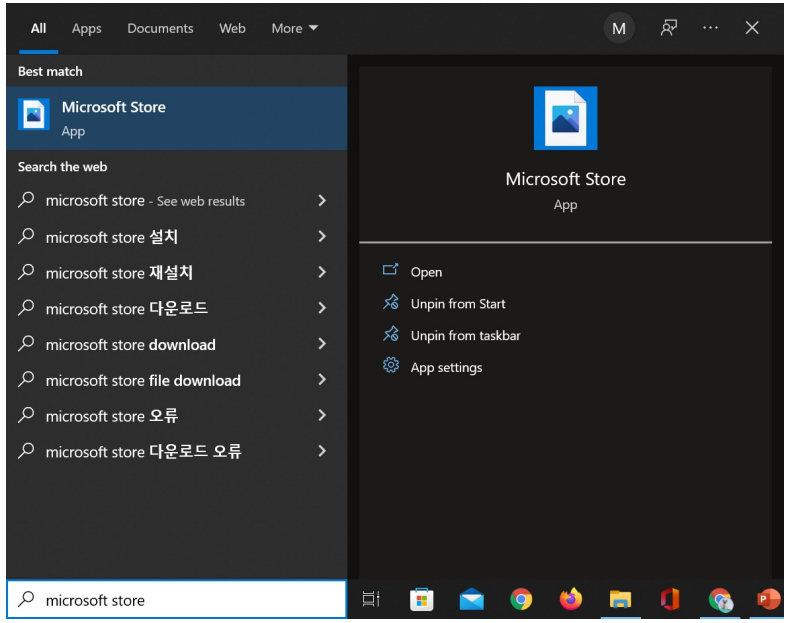
- Programs → Turn Windows features on or off → Select checkbox “Windows Subsystem for Linux” → OK → Restart

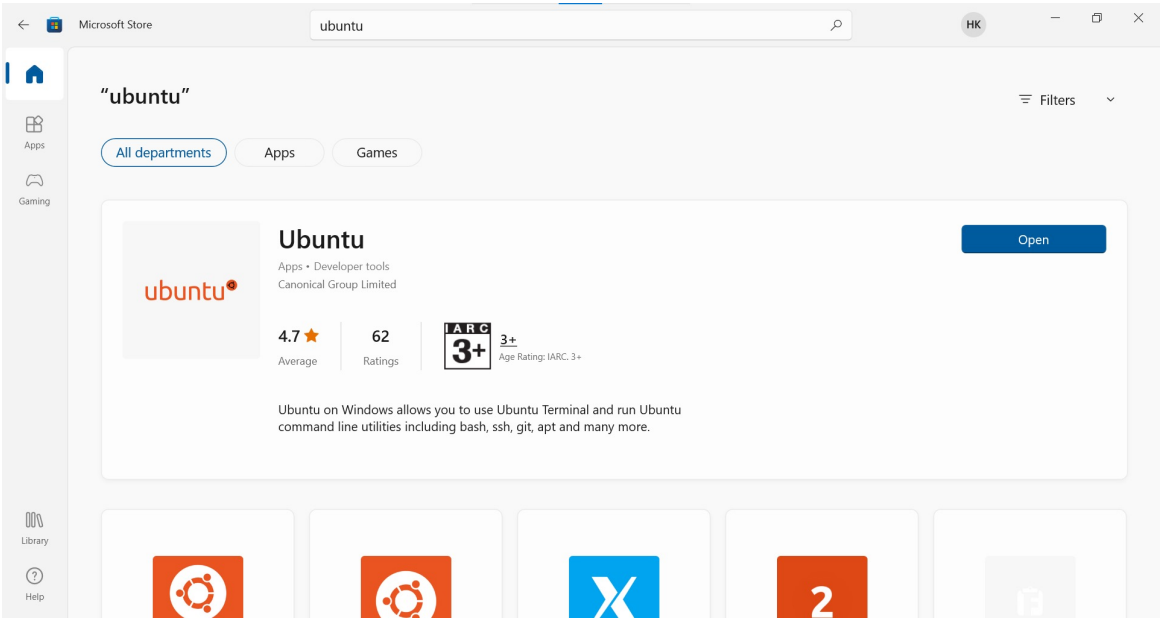




Installing Ubuntu in Windows 10

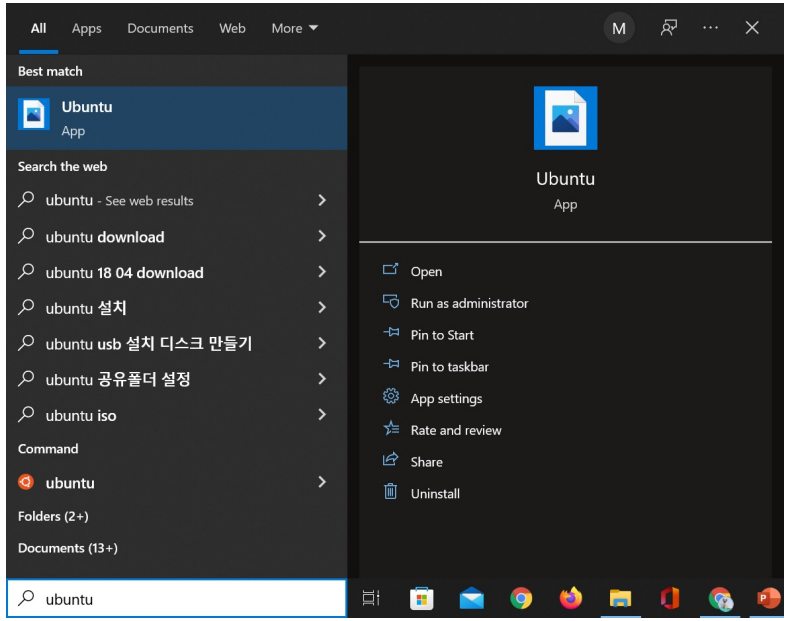
- Start menu → Microsoft Store → Search for “ubuntu” → Get Ubuntu

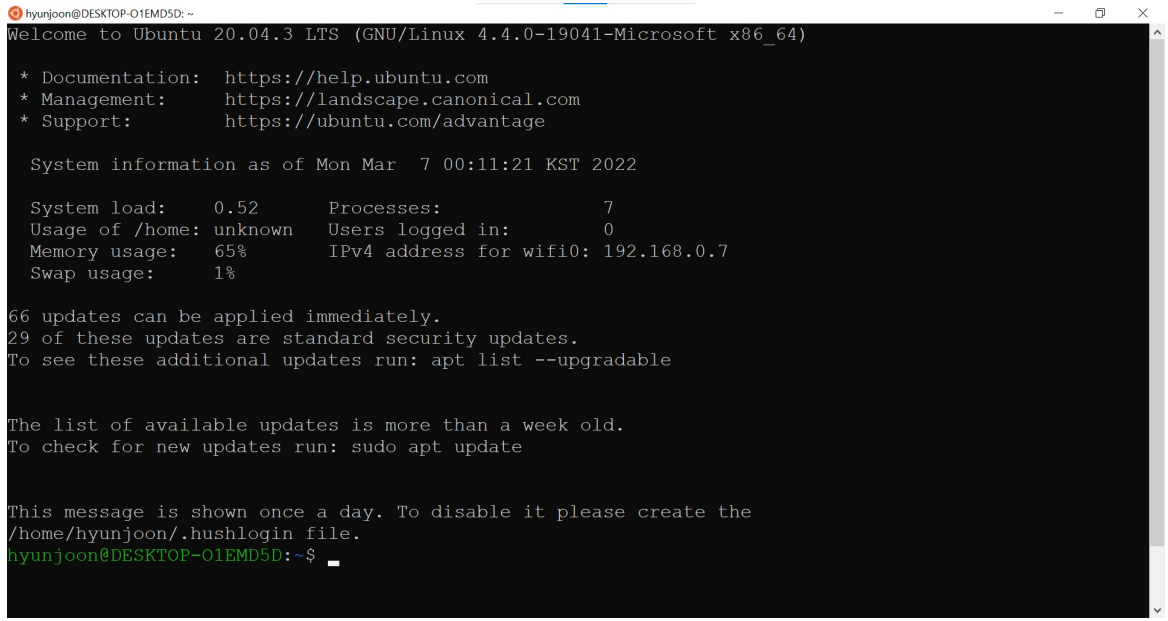




Installing Ubuntu in Windows 10

- Start menu → Ubuntu





Moving to the Directories in Windows

- Your drives are located in /mnt/
- cd /mnt/

hyunjoon@DESKTOP-O1EMD5D: /mnt/c/Users/Hyunjoon/Desktop

```
hyunjoon@DESKTOP-O1EMD5D:~$ ls /mnt/
c  g
hyunjoon@DESKTOP-O1EMD5D:~$ cd /mnt/c/
$Recycle.Bin/              OneDriveTemp/              System Volume Information/
$WinREAgent/               PerfLogs/                  Temp/
Config.Msi/                Program Files/             Users/
Documents and Settings/    Program Files (x86)/       Windows/
GPKI/                      ProgramData/               XecureSSL/
Intel/                     Recovery/                  system.sav/
Kings/                     SWsetup/
MozillaProfile/            SYS/
hyunjoon@DESKTOP-O1EMD5D:~$ cd /mnt/c/Users/Hyunjoon/Desktop/
hyunjoon@DESKTOP-O1EMD5D:/mnt/c/Users/Hyunjoon/Desktop$
```

Installing Ubuntu in Windows 11

- How can you fix Windows 11's WSL error
  - https://windowsreport.com/windows-11-wsl-error/

Linux Commands

Linux Commands

- Shell interprets and runs the below commands you type.

Command	Description	Command	Description
pwd	Current working directory	cat	Show the content of a file
ls	View the contents of a directory	diff	Compare the contents of two files
cd	Change directory	find	Search for a file or a directory
mkdir	Make a new directory	grep	Search for a keyword in a file
rmdir	Remove a directory	clear	Clean out the terminal
cp	Copy files		
rm	Remove files		
mv	Move (rename) files		

pwd

- Print the current working directory
- Return an absolute (full) path
  - absolute path: a path of all the directories that starts with "/".

```
[hjkim@localhost ~]$ pwd
/home/hjkim
```

ls

- View the contents of a directory
  - **`ls`**: current directory
  - **`ls /home/username/Documents`**: Documents directory
  - **`ls -a`**: show the hidden files. A hidden file starts with "."
  - **`ls -l`**: show the detailed information of files and directories.
    "**d**" stands for directory in **`drwxr-xr-x`**.
  - **`ls -al`**: **`ls -a`** + **`ls -l`**
  - **`ls -R`**: list all the files in the sub-directories as well

```
[hjkim@localhost /]$ ls
bin   data  etc   lib    media  opt   root  sbin   srv  tmp   usr
boot  dev   home  lib64  mnt    proc  run   share  sys  user  var
```

mkdir, rmdir, cd

- **`mkdir [directory_name]`**: create a new directory
  - `mkdir practice1`
- **`rmdir [directory_name]`**: remove an <u>empty</u> directory
  - `rmdir practice1`
- **`cd [directory_name]`**: move to [directory_name]
  - `cd practice1`
  - **`cd ..`** (with two dots) move to one directory up
  - **`cd -`** (with a hyphen) move to your previous directory

```
[hjkim@localhost /]$ ls
bin   data  etc   lib    media  opt   root  sbin   srv  tmp   usr
boot  dev   home  lib64  mnt    proc  run   share  sys  user  var
[hjkim@localhost /]$ cd data
[hjkim@localhost data]$ ls
continuous  dogs_vs_cats  prof    samba   swsong   yhnam
data        gmgu_data     rcheol  sghong  twitter
```

cp, rm, mv

- **`cp [curr] [another]`**: copy current file to another file
  - `cp practice1.txt practice1_copy.txt`
  - `cp /home/username/test.txt ../../`
- **`rm [target]`**: remove the target file
  - `rm practice1.txt`
- **`mv [target] [renamed_file/directory]`**: rename the target file to renamed file, or move the target file to another directory
  - `mv practice1.txt p1.txt`
  - `mv p1.txt ..`

cat, diff

- **cat**: list the contents of a file on the standard output
  - **cat file.txt**
- **diff**: compare the contents of two files line by line
  - **diff file1.txt file2.txt**

```
[hjkim@localhost ~]$ cat file1.txt
Hello
World
[hjkim@localhost ~]$ cat file2.txt
Wonderful
World
[hjkim@localhost ~]$ diff file1.txt file2.txt
1c1
< Hello
---
> Wonderful
```

find, grep

- **`find`**: search for a file or a directory
  - **`find /home/ -name file.txt`**: find file.txt in /home/
  - **`find . -name file.txt`**: find file.txt in the current directory
- **`grep`**: search all the texts in a given file for a given keyword
  - **`grep hello file.txt`**: find the word hello in file.txt

```
[hjkim@localhost ~]$ find . -name file1.txt
./file1.txt
```

```
[hjkim@localhost ~]$ grep -rn Hello file1.txt
1:Hello
```

clear

- **`clear`**: clean out the terminal

```
hyunjoon@DESKTOP-O1EMD5D: ~
hyunjoon@DESKTOP-O1EMD5D:~$ ls
hyunjoon@DESKTOP-O1EMD5D:~$ pwd
/home/hyunjoon
hyunjoon@DESKTOP-O1EMD5D:~$ echo "Hello world"
Hello world
hyunjoon@DESKTOP-O1EMD5D:~$ ls /mnt/
c/ g/
hyunjoon@DESKTOP-O1EMD5D:~$ ls /mnt/
c/ g/
hyunjoon@DESKTOP-O1EMD5D:~$ ls /mnt/c/
$Recycle.Bin/              PerfLogs/                  Windows/
$WinREAgent/               Program Files/             XecureSSL/
Config.Msi/                Program Files (x86)/       agentlog.txt
Documents and Settings/    ProgramData/               dism.log
DumpStack.log.tmp          Recovery/                  hiberfil.sys
GPKI/                      SWsetup/                   pagefile.sys
Intel/                     SYS/                       swapfile.sys
Kings/                     System Volume Information/ system.sav/
MozillaProfile/            Temp/                      tidylog.txt
OneDriveTemp/              Users/
hyunjoon@DESKTOP-O1EMD5D:~$ clear
```

```
hyunjoon@DESKTOP-O1EMD5D: ~
hyunjoon@DESKTOP-O1EMD5D:~$
```

Other Linux Commands

- Ctrl+C and Ctrl+Z are used to stop any command that is currently working.
  - Ctrl+C will stop and terminate the command
  - Ctrl+Z will simply pause the command.
- References
  - https://ubuntu.com/tutorials/command-line-for-beginners#3-opening-a-terminal
  - https://www.hostinger.com/tutorials/linux-commands
  - https://www.mireene.com/webimg/linux_tip1.htm
  - https://ss64.com/bash/

Vim

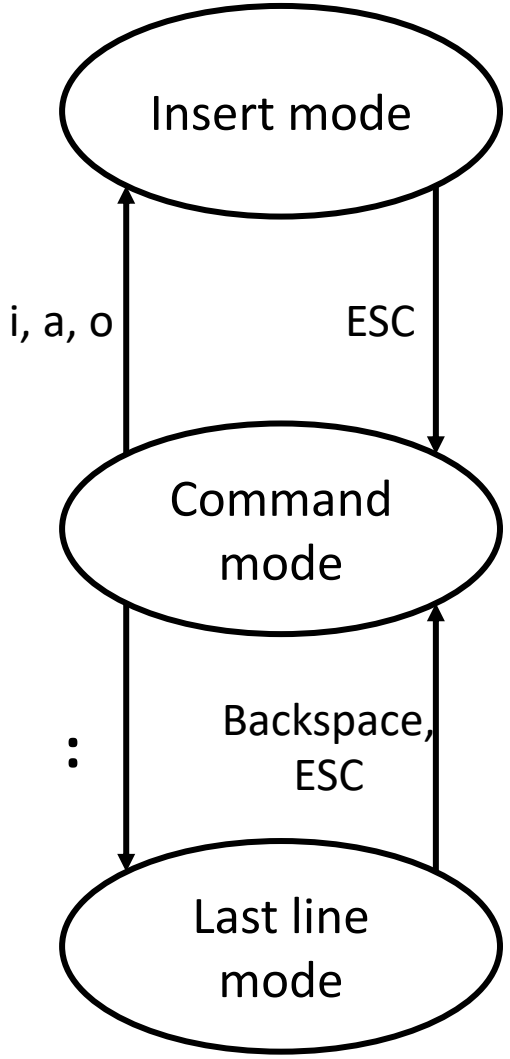
Usage of the Vi (or Vim) editor

Installation and Execution

- Type “vi” or “vim” in the terminal.
- If no Vim exists, install Vim.
  - **sudo apt-get update**
  - **sudo apt-get upgrade**
  - **sudo apt-get install vim**

Three Modes of Vim

- Command mode
  - When you start editing a file with Vim.
  - Issue commands (insert, append, delete, search, open, etc)
- Insert mode
  - Once you issue the insert, append or open commands.
  - Following text on the last line of your editor window indicates this mode.
    - --INSERT--
  - Type text into your file, and use the arrow keys to navigate around
  - Press [ESC] to move back to command mode
- Last line mode
  - Press ":" to get to this mode from command mode.
  - Search/replace or save/exit
  - Press [Backspace] or [ESC] to move back to command mode



Vimrc

- Customize the setting for your own Vim editor.
  - **vi ~/.vimrc**
- https://blog.outsider.ne.kr/518

```
syntax on              "Highlight syntax"
set number             "Present the row# where cursor locates"
set autoindent         "Automatic indentation"
set smartindent        "Smart indentation"
set hlsearch           "Highlight a search keyword"
set incsearch          "Incremental search when keyword is entered"
set backspace=eol,start,indent  "Backspace works at EOL, start, indent"
set expandtab          "Space instead of the tab key"
set tabstop=4          "Set the tab as 4 spaces"
set shiftwidth=2       "Automatic indentation of 2 spaces"
set smarttab           "Smart tab"
highlight Comment ctermfg=green
set history=1000
set visualbell
```

Vim Practice

- Write source code using Vim.

1. In terminal
   - **vi hello.cpp**
2. Press “i”, and write the code.
3. Press [ESC] to escape from insert mode.
4. Press “:wq” to save your file and exit.
   - **:w** write (save) into the file
   - **:q**quit

```
#include <iostream>
int main(void){
  std::cout<<"Hello world"<<std::endl;
  return 0;
}
```

Other Vim Commands

- References
  - https://www.guru99.com/the-vi-editor.html
  - https://blockdmask.tistory.com/25
  - https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&ved=2ahUKEwj7gNDs_oP0AhWJklYBHXUtADIQFnoECDYQAQ&url=http%3A%2F%2Fwww.atmos.albany.edu%2Fdaes%2Fatmclasses%2Fatm350%2Fvi_cheat_sheet.pdf&usg=AOvVaw3rx5RsHQVZ_PtrdR1Etg-H

Compilation and Execution

Compiling source code by using GCC

Running C/C++ and Python code

Compiling & Running C/C++ Code

- After writing the right code

1. Make sure your code is saved in the current directory.
   - **ls**
2. Compile
   - **g++ -std=c++11 hello.cpp**
   - **gcc hello.cpp**
3. Execute
   - **./a.out**

```
#include <cstdio>
#include <iostream>
int main(void){
  std::cout<<"Hello world"<<std::endl;
  return 0;
}
```

Compiling & Running C/C++ Code

- If you want to compile the source code into an executable file with another name
  - **g++ hello.cpp –o hello**
- Execute
  - **./hello**

Running Python Code

- Execute
  - **python3 hello.py**

```
def hello():
  print("Hello world"\n)

if __name__ == '__main__':
  hello()
```

File I/O Practice

- For each line of an input file:
  - Stock ticker of a company, its stock price, the number of its shares
  - Separated by a space.
  - **AAPL 166.23 16319441000**
- TODO: read the input file, and write an output file in which each line contains a company's stock ticker and total market cap
  - Separated by a space
  - Total market cap = stock price × number of shares
  - **AAPL 2712780677430.0**

File I/O Practice Examples

- C/C++
  - g++ io.cpp
  - ./a.out input.txt output.txt
- Python
  - python3 io.py input.txt output.txt

```
AAPL 166.23 16319441000
MSFT 295.92 7496866428
GOOGL 2677.99 660971321
AMZN 2957.97 508844410
FB 202.97 2721941960
TSLA 839.29 1033507611
```

input.txt

```
AAPL 2712780677430.0
MSFT 2218472713373.7603
GOOGL 1770074587924.7898
AMZN 1505146499447.7
FB 552472559621.2
TSLA 867412602836.19
```

output.txt